



East West University

Department of Computer Science and Engineering

Semester : Spring 2021

Course Title : Electrical Circuits

Course Code : CSE 209

Section : 03

Experiment Number : 05

Experiment Title : Verification of Superposition Theorem

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Date of Performance: 07.04.2021

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Objectives:

1. To verify the superposition theorem theoretically, experimentally, and using PSpice simulation.

Circuit Diagram:

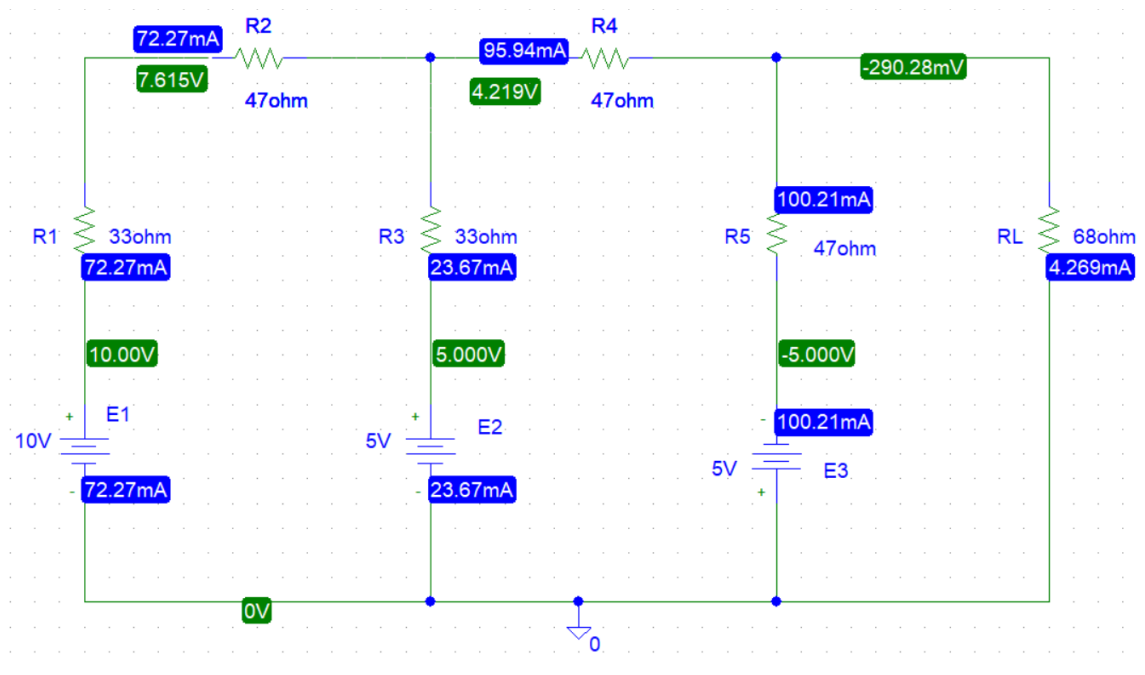


Figure 1. Circuit with all sources active.

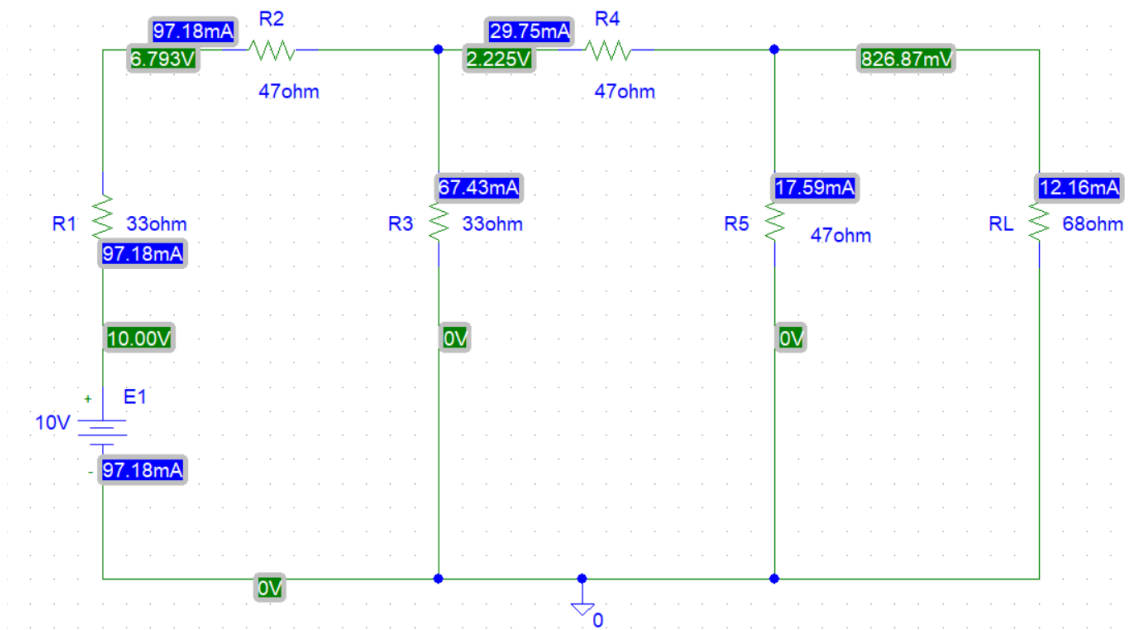


Figure 2. Circuit with E_1 source active.

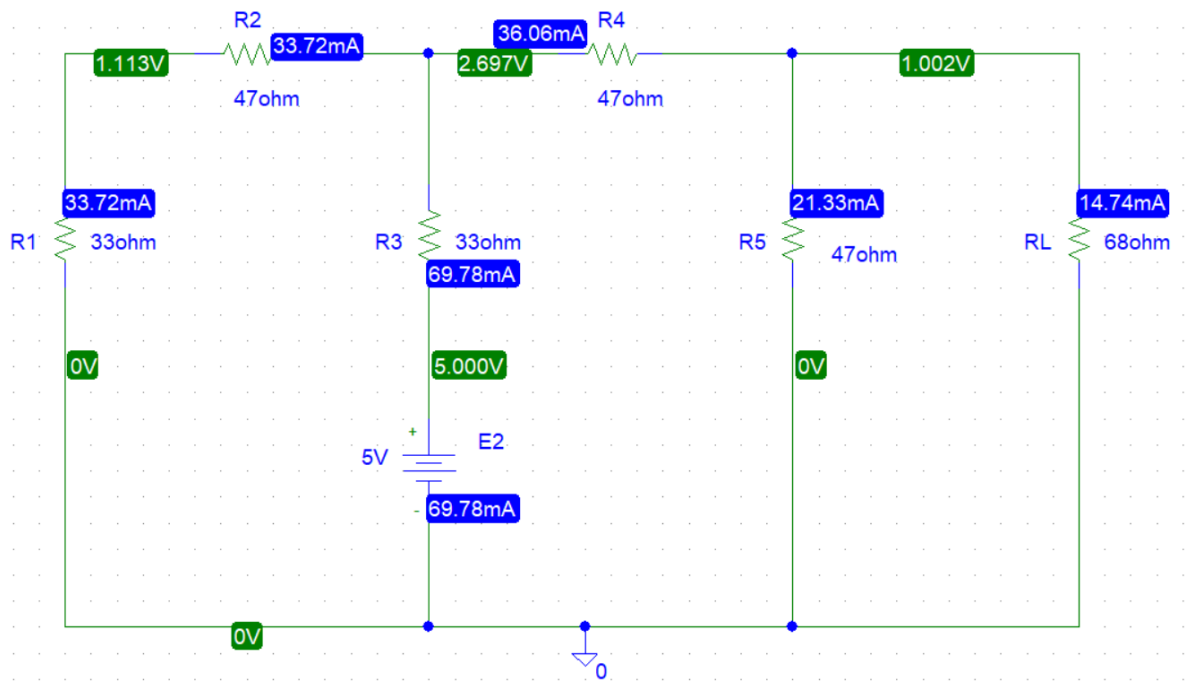


Figure 3. Circuit with E_2 source active.

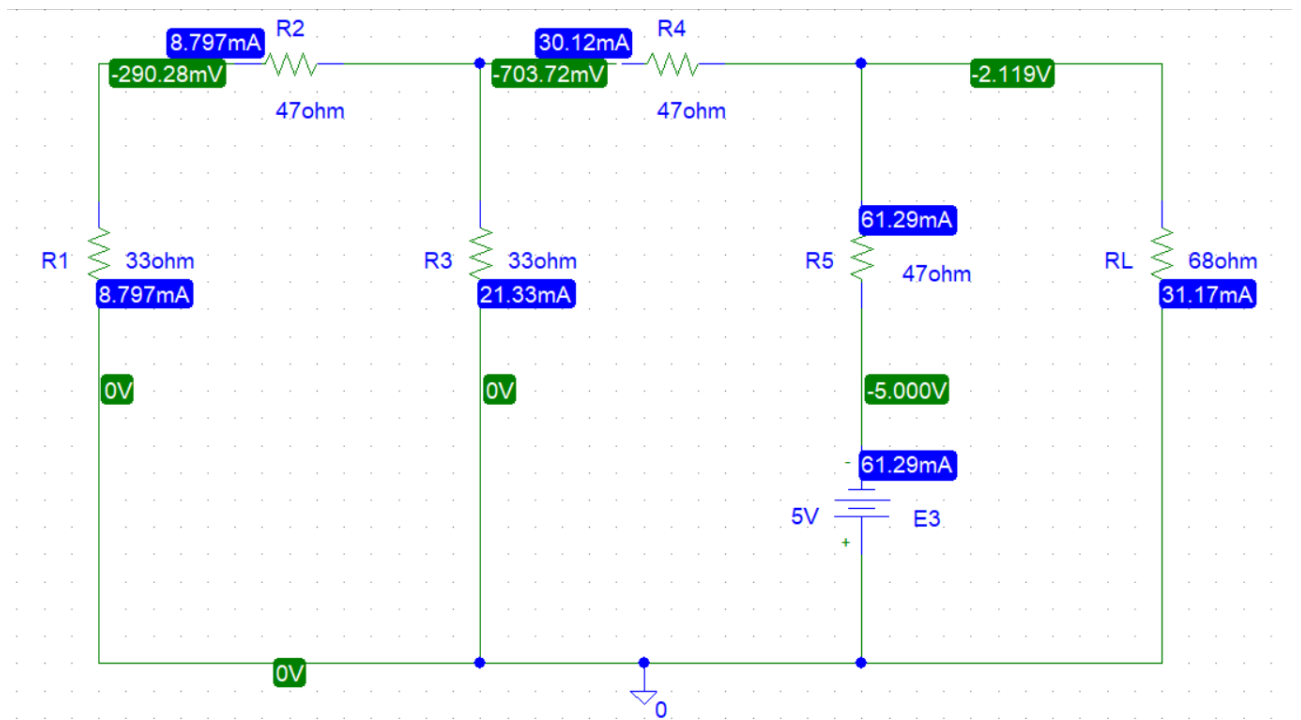


Figure 4. Circuit with E_3 source active.

Experimental Datasheet:

Value of E_1 (V)	Value of E_2 (V)	Value of E_3 (V)	Value of I_L with all sources active(mA)	Value of I_{L1} with only E_1 active(mA)	Value of I_{L2} with only E_2 active(mA)	Value of I_{L3} with only E_2 active (mA)	Value of Resistors(Ω)
10	5	-5	-4.269	12.16	14.74	-31.17	$R_1 = 33\Omega$ $R_2 = 47\Omega$ $R_3 = 33\Omega$ $R_4 = 47\Omega$ $R_5 = 47\Omega$ $R_L = 68\Omega$

Answers to the post lab report questions:

1. For figure 1:

Applying KVL at mesh 1,

$$33 i_1 + 47 i_1 + 33(i_1 - i_2) = 10 - 5$$

$$33 i_1 + 47 i_1 + 33 i_1 - 33 i_2 = 5$$

$$113 i_1 - 33 i_2 + 0 i_3 = 5 \dots\dots (1)$$

Applying KVL at mesh 2,

$$33(i_2 - i_1) + 47 i_2 + 47(i_2 - i_3) = 5 + 5$$

$$33 i_2 - 33 i_1 + 47 i_2 + 47 i_2 - 47 i_3 = 10$$

$$-33 i_1 + 127 i_2 - 47 i_3 = 10 \dots\dots (2)$$

Applying KVL at mesh 3,

$$47(i_3 - i_2) + 68i_3 = -5$$

$$47i_3 - 47i_2 + 68i_3 = -5$$

$$0i_1 - 47i_2 + 115i_3 = -5 \dots\dots(3)$$

Solving equation 1, 2 & 3,

$$i_1 = 0.07227\text{A} = 72.27\text{mA}$$

$$i_2 = 0.09594\text{A} = 95.94\text{mA}$$

$$i_3 = -4.269 \times 10^{-3}\text{A} = -4.269\text{mA}$$

$$\text{here, } i_3 = I_L = -4.269\text{mA}$$

For figure 2:

Applying KVL at mesh 1,

$$33i_1 + 47i_1 + 33(i_1 - i_2) = 10$$

$$33i_1 + 47i_1 + 33i_1 - 33i_2 = 10$$

$$113i_1 - 33i_2 + 0i_3 = 10 \dots\dots(1)$$

Applying KVL at mesh 2,

$$33(i_2 - i_1) + 47i_2 + 47(i_2 - i_3) = 0$$

$$33i_2 - 33i_1 + 47i_2 + 47i_2 - 47i_3 = 0$$

$$-33i_1 + 127i_2 - 47i_3 = 0 \dots\dots(2)$$

Applying KVL at mesh 3,

$$47(i_3 - i_2) + 68i_3 = 0$$

$$47i_3 - 47i_2 + 68i_3 = 0$$

$$0i_1 - 47i_2 + 115i_3 = 0 \dots\dots(3)$$

Solving equation 1, 2 & 3,

$$i_1 = 0.09718\text{A} = 97.18\text{mA}$$

$$i_2 = 0.02975\text{A} = 29.75\text{mA}$$

$$i_3 = 0.01216\text{A} = 12.16\text{mA}$$

$$\text{here, } i_3 = I_{L1} = 12.16\text{mA}$$

For figure 3:

Applying KVL at mesh 1,

$$33i_1 + 47i_1 + 33(i_1 - i_2) = -5$$

$$33i_1 + 47i_1 + 33i_1 - 33i_2 = -5$$

$$113i_1 - 33i_2 + 0i_3 = -5 \dots\dots(1)$$

Applying KVL at mesh 2,

$$33(i_2 - i_1) + 47i_2 + 47(i_2 - i_3) = 5$$

$$33i_2 - 33i_1 + 47i_2 + 47i_2 - 47i_3 = 5$$

$$-33i_1 + 127i_2 - 47i_3 = 5 \dots\dots(2)$$

Applying KVL at mesh 3,

$$47(i_3 - i_2) + 68i_3 = 0$$

$$47i_3 - 47i_2 + 68i_3 = 0$$

$$0i_1 - 47i_2 + 115i_3 = 0 \dots\dots(3)$$

Solving equation 1, 2 & 3,

$$i_1 = -0.03372\text{A} = -33.72\text{mA}$$

$$i_2 = 0.03606\text{A} = 36.06\text{mA}$$

$$i_3 = 0.01474\text{A} = 14.74\text{mA}$$

$$\text{here, } i_3 = I_{L2} = 14.74\text{mA}$$

For figure 4:

Applying KVL at mesh 1,

$$33 i_1 + 47 i_1 + 33(i_1 - i_2) = 0$$

$$33 i_1 + 47 i_1 + 33 i_1 - 33 i_2 = 0$$

$$113 i_1 - 33 i_2 + 0 i_3 = 0 \dots\dots(1)$$

Applying KVL at mesh 2,

$$33(i_2 - i_1) + 47 i_2 + 47(i_2 - i_3) = 5$$

$$33 i_2 - 33 i_1 + 47 i_2 + 47 i_2 - 47 i_3 = 5$$

$$-33 i_1 + 127 i_2 - 47 i_3 = 5 \dots\dots(2)$$

Applying KVL at mesh 3,

$$47(i_3 - i_2) + 68 i_3 = -5$$

$$47 i_3 - 47 i_2 + 68 i_3 = -5$$

$$0 i_1 - 47 i_2 + 115 i_3 = -5 \dots\dots(3)$$

Solving equation 1, 2 & 3,

$$i_1 = 8.797 \times 10^{-3} \text{A} = 8.797 \text{mA}$$

$$i_2 = 0.03012 \text{A} = 30.12 \text{mA}$$

$$i_3 = -0.03117 \text{A} = -31.17 \text{mA}$$

$$\text{here, } i_3 = I_{L3} = -31.17 \text{mA}$$

Solving the equations we get,

$$I_L = -4.269 \text{mA} = -4.27 \text{mA}$$

$$I_{L1} = 12.16 \text{mA}$$

$$I_{L2} = 14.74 \text{mA}$$

$$I_{L3} = -31.17 \text{mA}$$

$$I_L = I_{L1} + I_{L2} + I_{L3}$$

$$=12.16+14.74-31.17$$

$$= -4.27\text{mA}$$

It holds the superposition theorem.

As the experiment was done in PSpice, the values of current are same. Thus, there is no discrepancy.

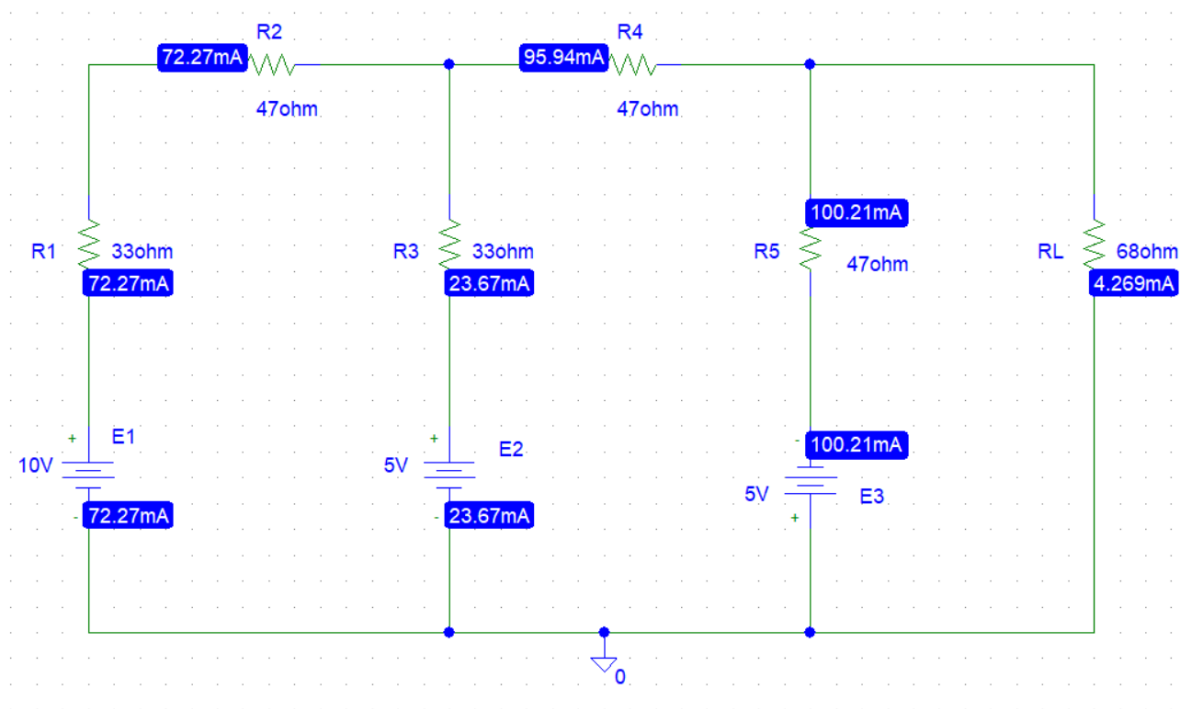


Figure 1. Circuit with all sources active.

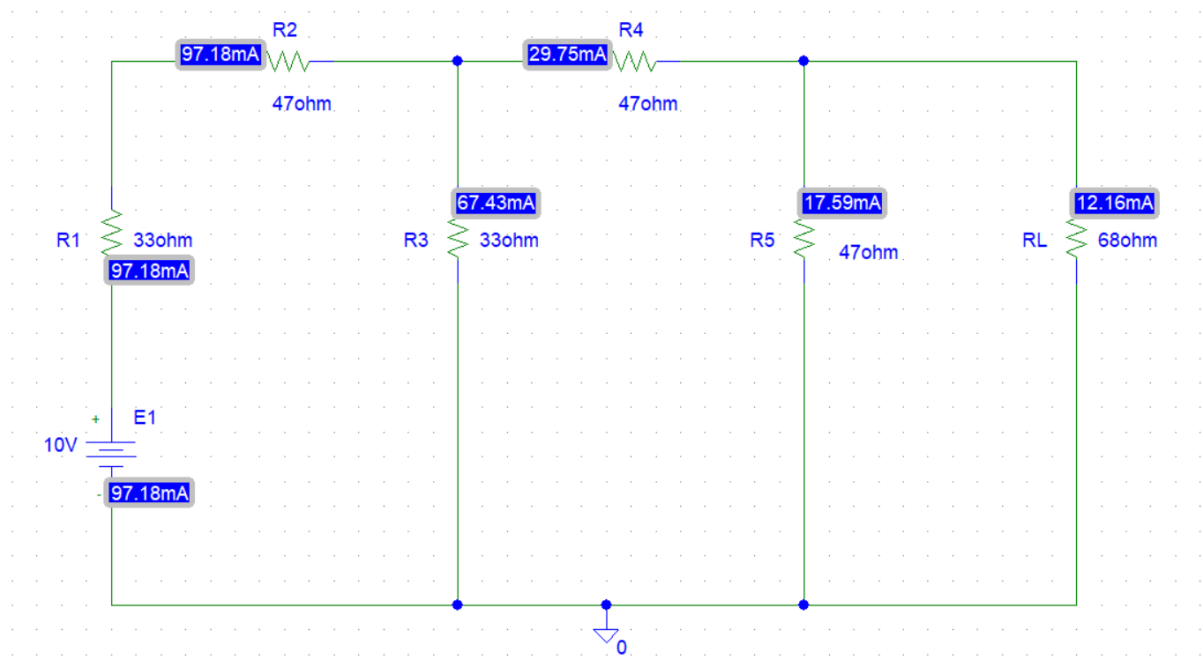


Figure 2. Circuit with E_1 source active.

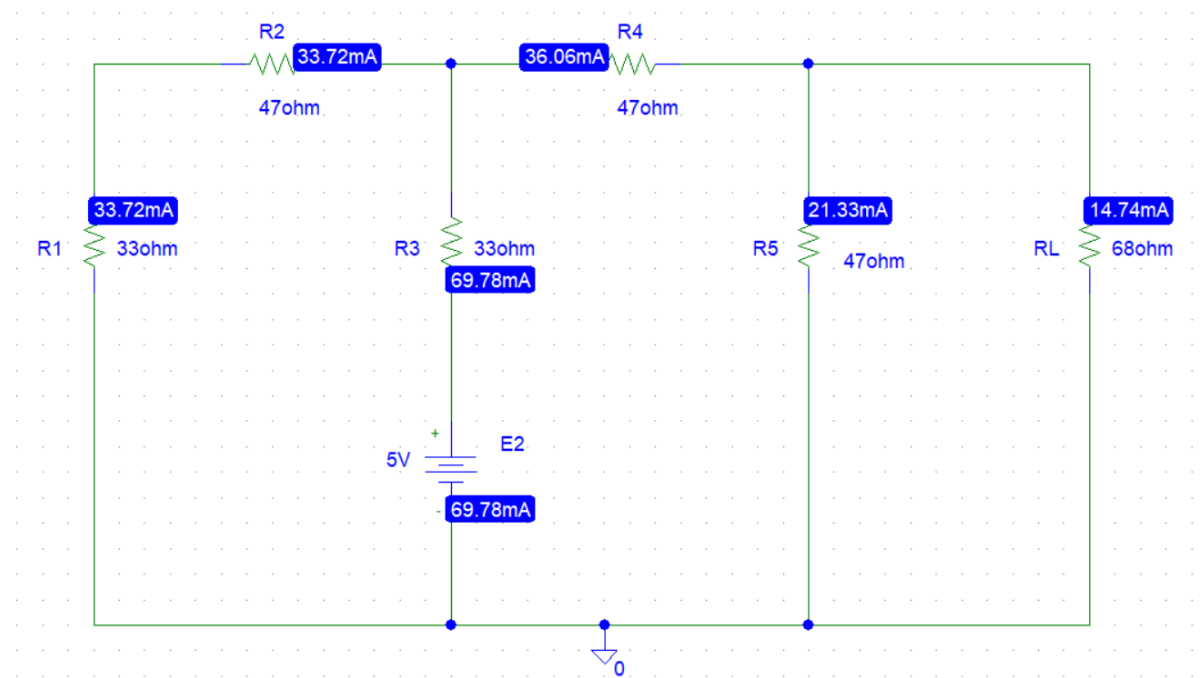


Figure 3. Circuit with E_2 source active.

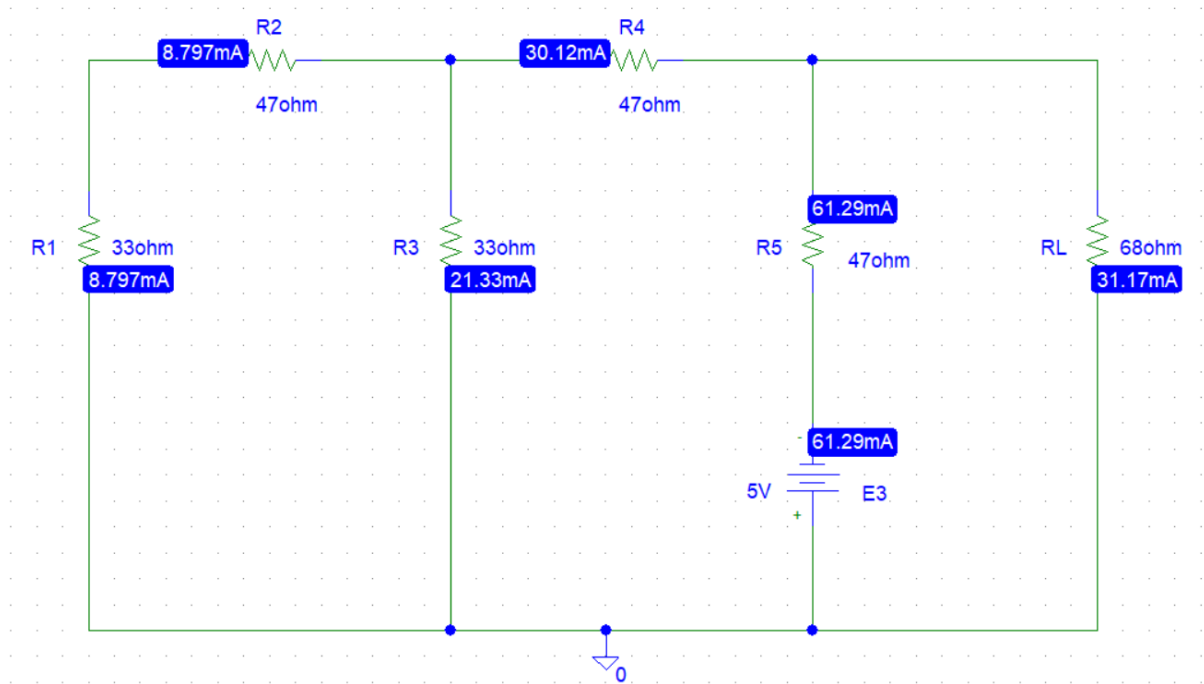


Figure 4. Circuit with E_3 source active.

Here ,

$$I_L = -4.269\text{mA} = -4.27\text{mA}$$

$$I_{L1} = 12.16\text{mA}$$

$$I_{L2} = 14.74\text{mA}$$

$$I_{L3} = -31.17\text{mA}$$

$$I_L = I_{L1} + I_{L2} + I_{L3}$$

$$= 12.16 + 14.74 - 31.17$$

$$= -4.27\text{mA}$$

It holds the superposition theorem.

As the experiment was done in PSpice, the theoretical values & the PSpice values of current are same. Thus , there is no discrepancy.

Conclusion:

From the experiment we learnt about the superposition theorem. At first we calculated I_L . Then we made independent sources inactive (only one at a time) and calculated I_{L1} , I_{L2} , I_{L3} . We found that the algebraic sum of I_{L1} , I_{L2} , I_{L3} is equal to I_L . Thus, Superposition theorem was verified. This experiment was done in PSpice so, the theoretical values & the PSpice values of current are same.